

Troubleshooting Hardware -- Guided Notes ANSWER KEY

Unit 4.2 | CompTIA Network+ N10-009 Obj. Core 2 5.2 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

POST stands for _____. It runs every time a computer _____ and tests the CPU, RAM, storage, and other hardware before loading the OS. If POST fails, the system will _____ with an error code, beep code, or stop code displayed on screen.

Answer: Power-On Self-Test; powers on (boots); halt (stop).

Question 2

Beep codes are audio signals the motherboard emits when a POST failure occurs before video output is available. The meaning of beep codes varies by _____ manufacturer. One long beep followed by two short beeps on many systems indicates a _____ failure. To interpret a beep code, you should consult the motherboard's _____.

Answer: BIOS (motherboard); video (GPU / display); documentation (manual).

Question 3

When a system powers on but shows no display, the first two things to check are: (1) the _____ cable is firmly connected to the correct port (the _____ rather than the motherboard if a discrete GPU is installed), and (2) the _____ is fully seated in its PCIe slot. These steps resolve the majority of no-display issues.

Answer: Monitor (display); GPU (graphics card); GPU (graphics card).

Question 4

A _____ sound coming from a hard drive is a serious symptom indicating mechanical failure -- the read/write heads are likely making contact with the _____. The correct response is to _____ immediately before attempting any further diagnosis. Running a drive that is clicking risks permanent _____ loss.

Answer: Clicking (grinding); platters (disk surface); back up all data; data.

Question 5

Failing RAM typically causes _____ (random errors causing Windows to crash with a blue screen) and random restarts. RAM failure can also prevent POST entirely. To isolate failing RAM, a technician should test _____ stick at a time and run a tool such as _____.

Answer: BSODs (Blue Screens of Death); one; MemTest86.

Question 6

_____ are visual display errors such as random pixels, color blocks, or flickering -- they indicate the GPU may be overheating or failing. When the CPU reaches its maximum safe temperature, it reduces its clock speed in a process called _____. If the CPU temperature limit is exceeded, the

system will perform a _____ shutdown to prevent damage.

Answer: GPU artifacts (visual artifacts); thermal throttling; emergency (protective) shutdown.

Question 7

Visually inspecting a motherboard for _____ capacitors (bulging or leaking tops) is a reliable sign that the board needs to be replaced. A system that will not power on at all -- no fans, no LEDs, nothing -- should first have the technician check the _____ switch on the back of the PSU, which is sometimes accidentally switched to the _____ position.

Answer: Bulging (swollen / failed); rear power (PSU on/off); off (0).

Question 8 -- Real World Application

REAL WORLD: A user drops off a desktop that was working fine but now powers on, the CPU fan spins, and the keyboard lights flash -- but there is no video output at all and no beep codes. Describe your complete diagnostic process in the correct order, including what you would check, test, or swap at each step.

Answer: Step 1 -- Check the monitor and cable. Test with a known-good monitor and cable connected to the GPU output port (not motherboard). Step 2 -- Reseat the GPU in the PCIe x16 slot and confirm PCIe power connectors are attached. Step 3 -- If still no video, remove the GPU and test with integrated graphics (if CPU has it) connected to the motherboard display port. Step 4 -- Reseat RAM sticks. Try one stick at a time in slot A2 (typically the recommended single-stick slot). Step 5 -- Listen for beep codes -- if a speaker is not connected, connect one to the front panel header. Step 6 -- Clear CMOS by removing the battery for 30 seconds to reset BIOS to defaults. Step 7 -- If none of the above work, suspect a failed CPU, motherboard, or PSU and begin component swap testing.